

2022

Cybersecurity  
INSIDERS

# VULNERABILITY MANAGEMENT REPORT



  
helpsystems

# INTRODUCTION

In a rapidly changing threat environment, it becomes increasingly challenging for organizations to identify, prioritize, remediate or mitigate software and system vulnerabilities. This challenge is exacerbated by the continued shortage of skilled cybersecurity professionals.

In this context, cybersecurity teams need accessible, effective vulnerability management solutions that help scale security efforts to protect the business without adding additional staff. This also includes layered security solutions beyond vulnerability scanning that work in concert to improve team efficiency and security effectiveness.

The 2022 Vulnerability Management Report is based on a comprehensive survey of over 390 cybersecurity professionals in September 2022 to gain insights into the latest trends, key challenges, and solution preferences for vulnerability management.

## Key findings include:

- While a majority of organizations (71%) run a formal vulnerability management program in-house, only about a third consider their program very effective (30%).
- Budget constraints are the biggest barrier to better vulnerability management (60%), followed by the perennial skills shortage (45%) and ineffective vulnerability management processes (36%).
- IoT and OT devices (65%) top the list of infrastructure requiring better vulnerability management, followed by cloud assets (44%) and endpoints (41%).
- Only about a third of organizations deploy patches within three days of availability (29%) - another third take between three days and two weeks (35%), the rest up to a month or more.
- While most organizations experienced an increase in vulnerabilities over the past 12 months (76%), only about a third (30%) expect an increase in staffing for vulnerability management.
- The good news: 44% of organizations expect an increase in investment for vulnerability management solutions.
- When evaluating solutions, cybersecurity professionals prioritize accuracy of vulnerability detection (79%), followed by analytics features (63%) and cost of ownership (59%). Vulnerability assessment (70%) tops the list of required features, followed by asset discovery (66%), vulnerability scanning (63%), and risk management features (61%).

We would like to thank Digital Defense and Beyond Security by [HelpSystems](#), global leaders in vulnerability management solutions, for supporting this important research. We hope you find this report informative and helpful as you continue your efforts in protecting your organizations against evolving threats.

Thank you,

*Holger Schulze*



**Holger Schulze**

CEO and Founder  
Cybersecurity Insiders

**Cybersecurity**  
INSIDERS

# VULNERABILITY MANAGEMENT PROGRAM

We asked cybersecurity professionals whether their organization has a formal vulnerability management program in place. The good news: A significant majority of organizations (71%) run a formal vulnerability management program in-house. Few organizations have only informal, ad-hoc programs (12%), third-party managed programs (8%), or no program at all (9%).

## ► Does your organization have a vulnerability management program in place?



Yes, we have a formal program managed in-house



71%

Yes, we have a formal program managed by a third-party provider



8%

Yes, we have an informal, ad-hoc program



12%

No, we do not have a program yet

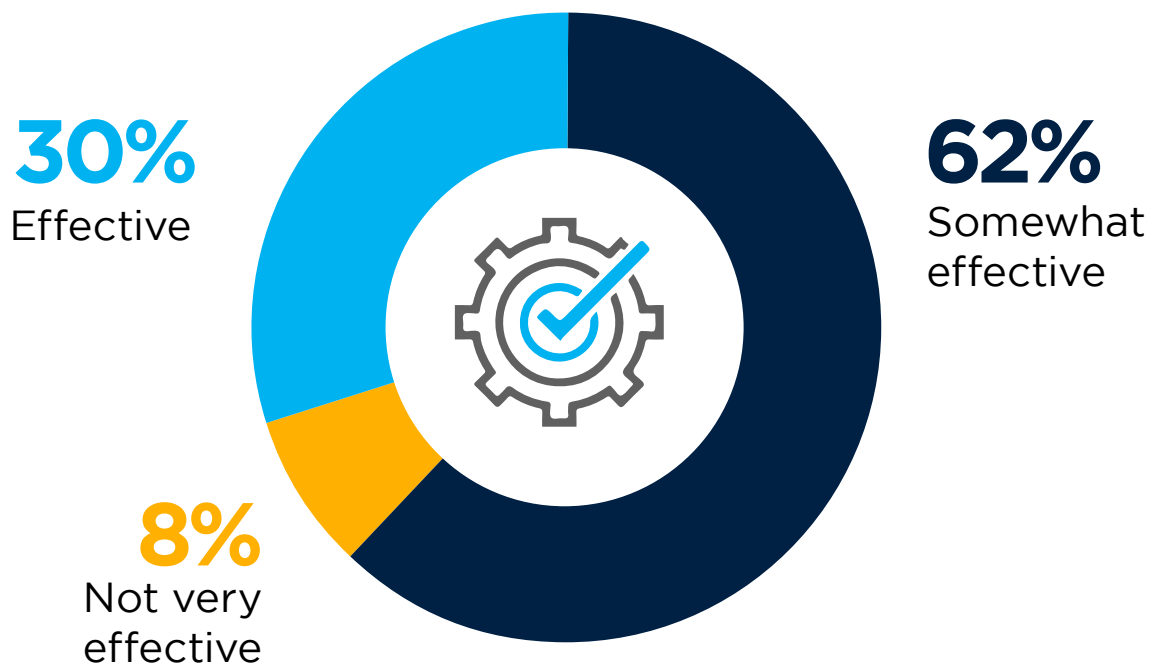


9%

# VULNERABILITY MANAGEMENT EFFECTIVENESS

When asked about the effectiveness of their vulnerability management programs, nearly a third of organizations consider their program effective (30%). For a majority, vulnerability management is only somewhat effective (62%).

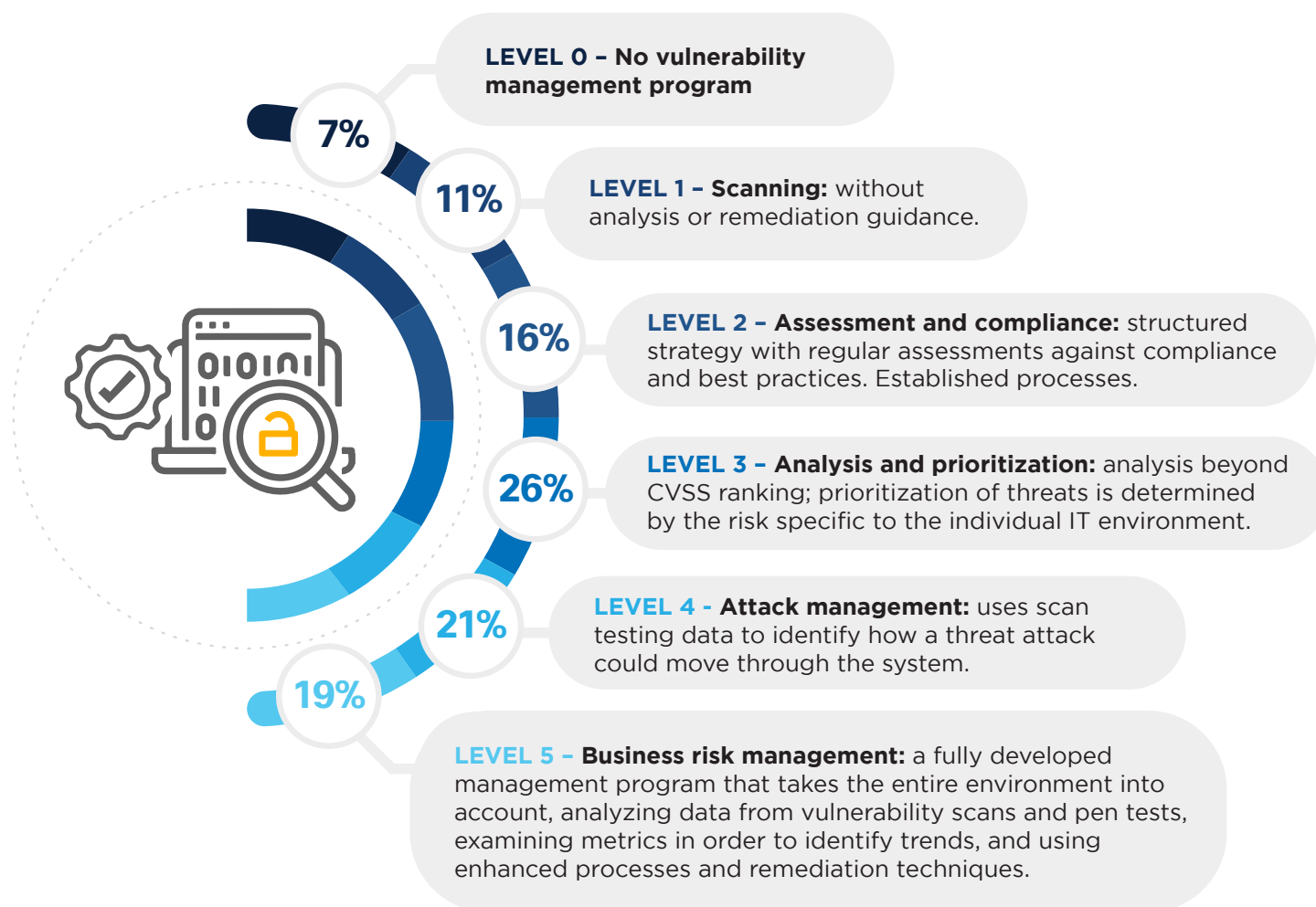
► How would you rate the effectiveness of your current vulnerability management program?



# VULNERABILITY MANAGEMENT MATURITY

Although 71% of organizations have a formal vulnerability management program in place, we wanted to drill down a bit deeper to assess the level of maturity and effectiveness those programs have reached. Most organizations (66%) have reached maturity Level 3 or higher. This level is characterized by analysis and prioritization of risk specific to the IT environment.

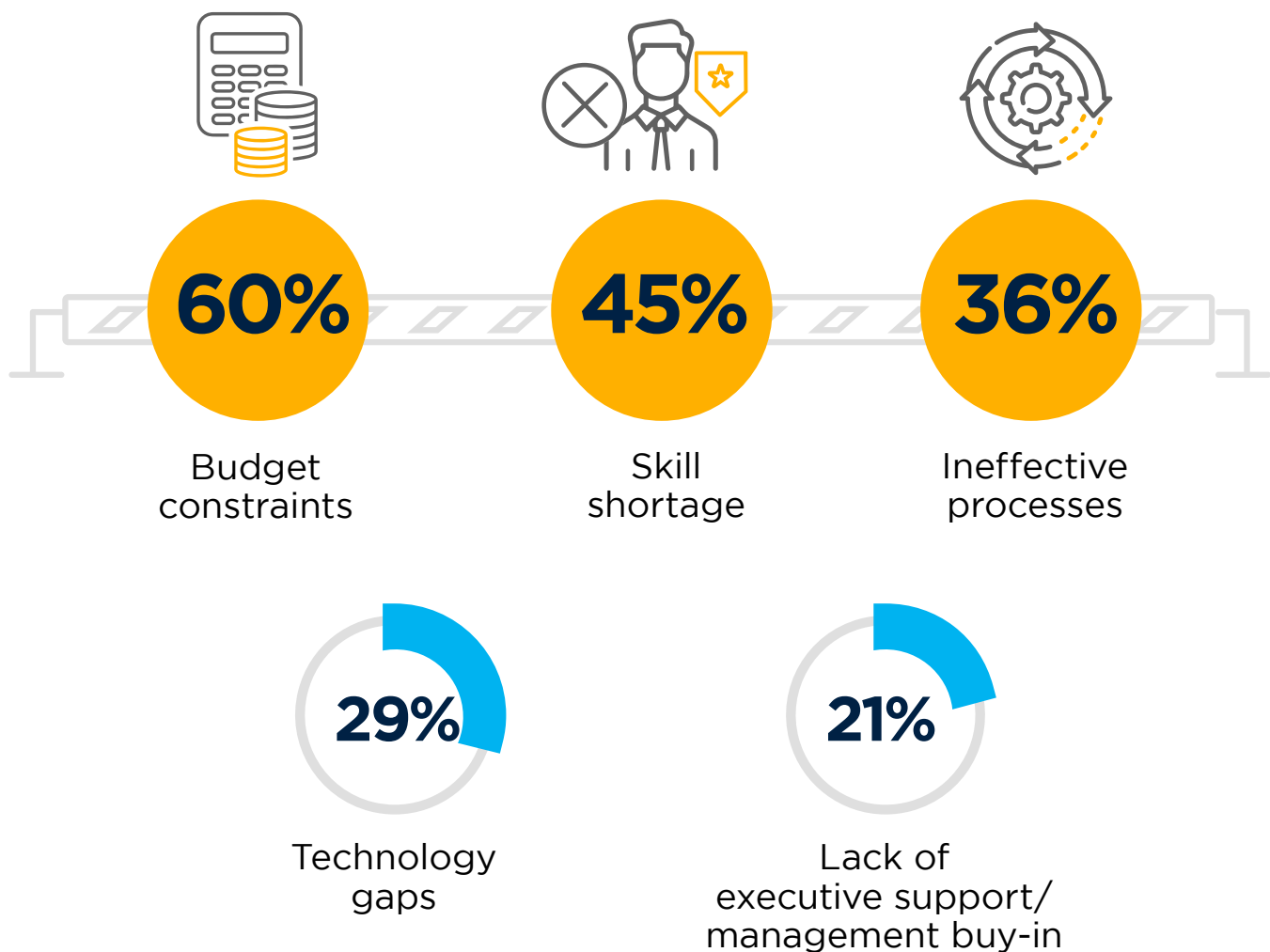
## ► How would you characterize the level of maturity your vulnerability management program has reached?



# BARRIERS TO BETTER VULNERABILITY MANAGEMENT

Budget constraints tops the list of biggest barriers to better vulnerability management (60%), followed at a distance by the perennial skills shortage (45%) and ineffective vulnerability management processes (36%). Lack of executive support (21%) and technology gaps (29%) play a smaller role as barriers.

► In your organization, what are the biggest barriers to better vulnerability management?

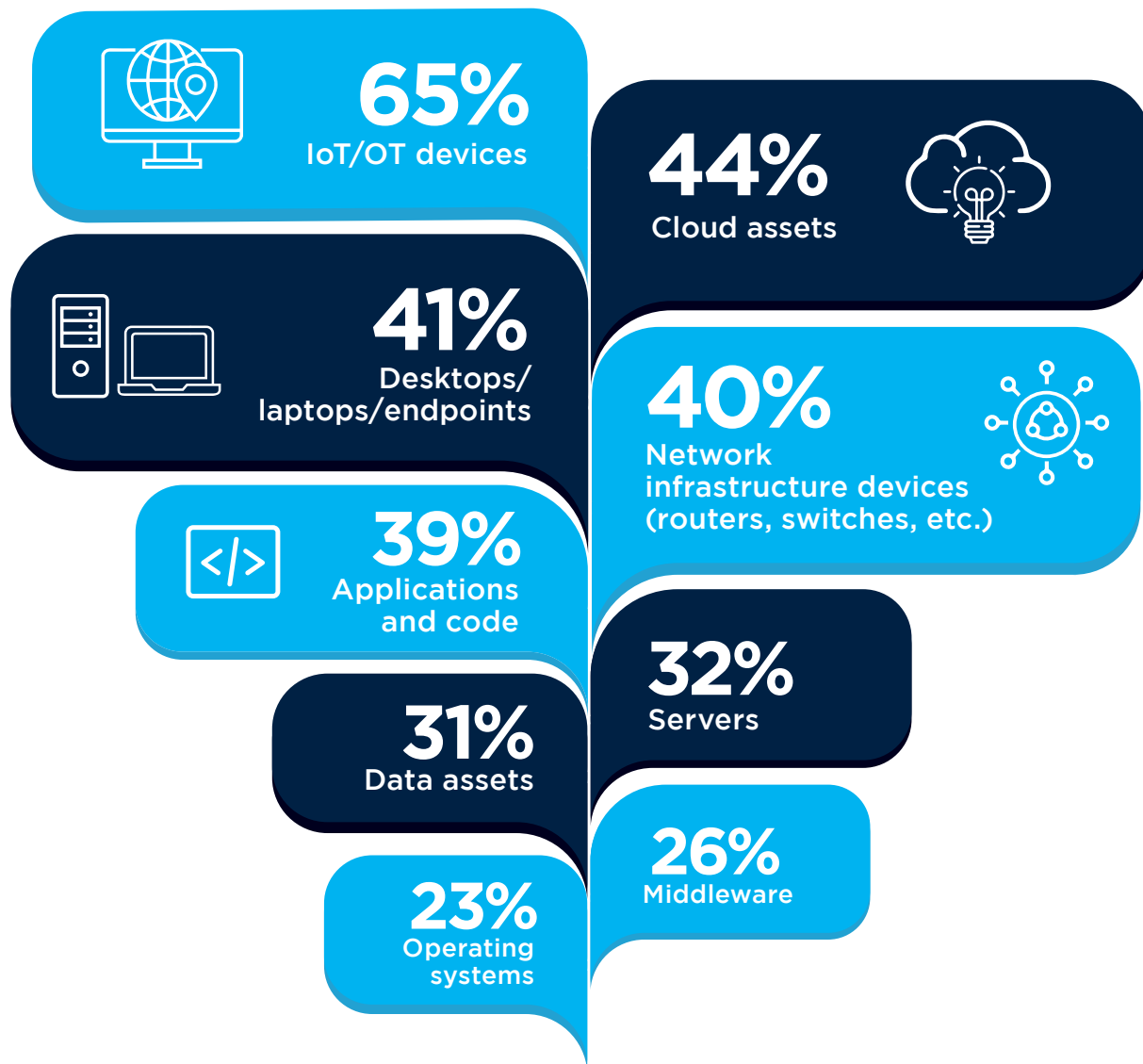


Other 11%

# FOCUS AREAS

We asked cybersecurity professionals which areas of their organization's infrastructure need better vulnerability management. IoT and OT devices tops the list at 65%, reflecting the neglect these devices experience in most IT environments, from a security perspective. This is followed by cloud assets (44%) and endpoints (41%).

## ▶ Which areas of your infrastructure do you think need better/more vulnerability management?



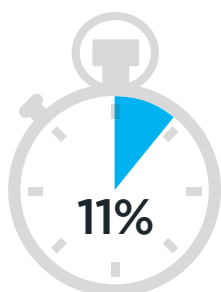
Other 6%

# SPEED OF SECURITY PATCHING

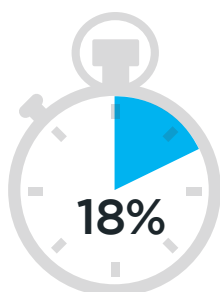
Speed of patching is critical in maintaining a robust security posture. How quickly do organizations deploy security patches to mitigate vulnerabilities? Just under a third of organizations deploy within three days of availability (29%). Twenty-two percent deploy patches within a week of availability, while another third take between one week and a month (35%).

► **Once available, on average, how long does it take to deploy a security patch in your environment?**

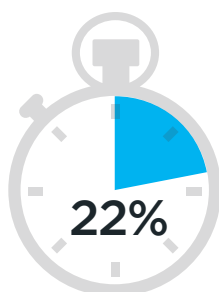
**35%**   
need between two  
weeks and a month to  
deploy a security patch



Same day



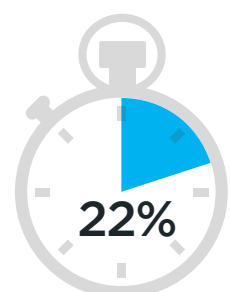
Within  
3 days



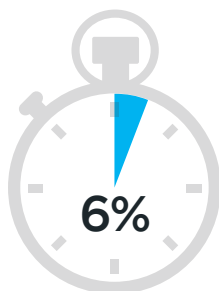
Within  
1 week



Within  
2 weeks



Within  
1 month



Within  
3 months



Within  
6 months



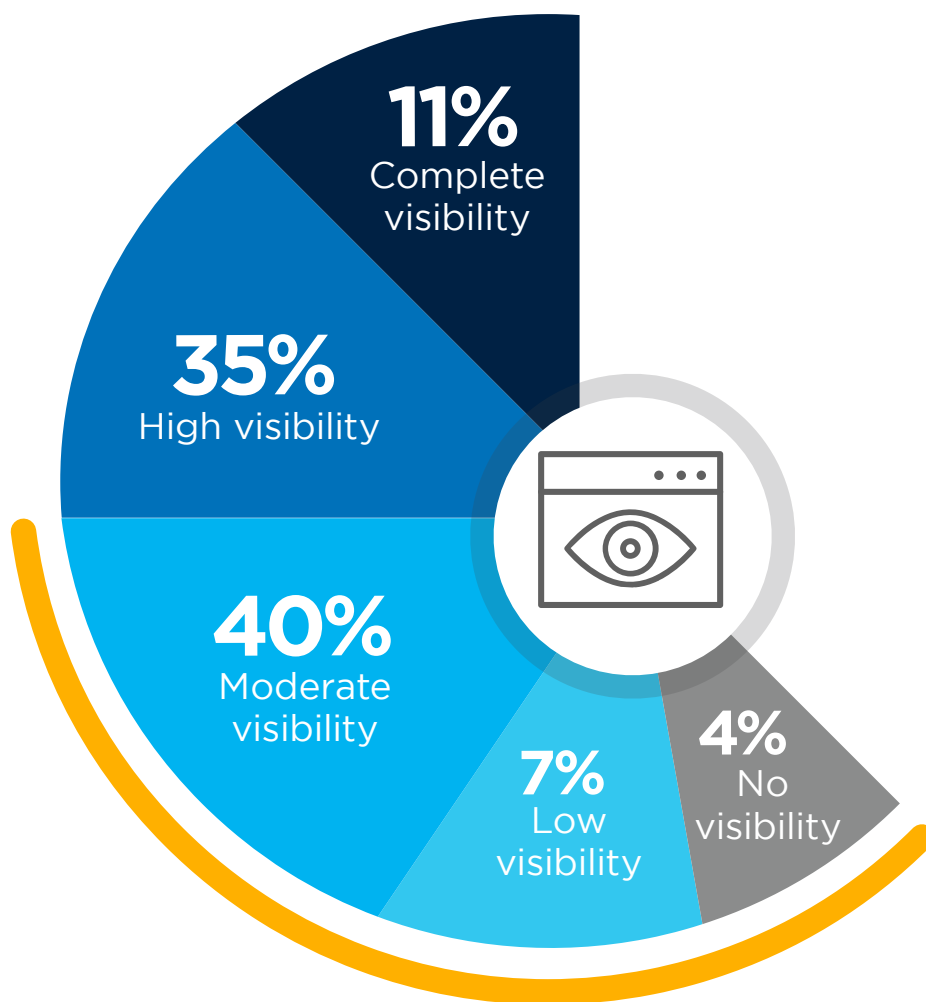
Within  
1 year or more



# LIMITED VISIBILITY INTO VULNERABILITIES

Visibility into vulnerabilities is the first step to addressing them. We asked cybersecurity professionals about their visibility into vulnerabilities, and less than half of them claim to have high (35%) or complete visibility (11%). Over half of organizations (51%) have only moderate visibility into their vulnerabilities, at best.

► **What level of visibility do you have into vulnerabilities across your IT environment?**



**Over half of organizations (51%), have at most moderate visibility into their vulnerabilities**

Unsure 3%

# VOLUME OF SECURITY VULNERABILITIES

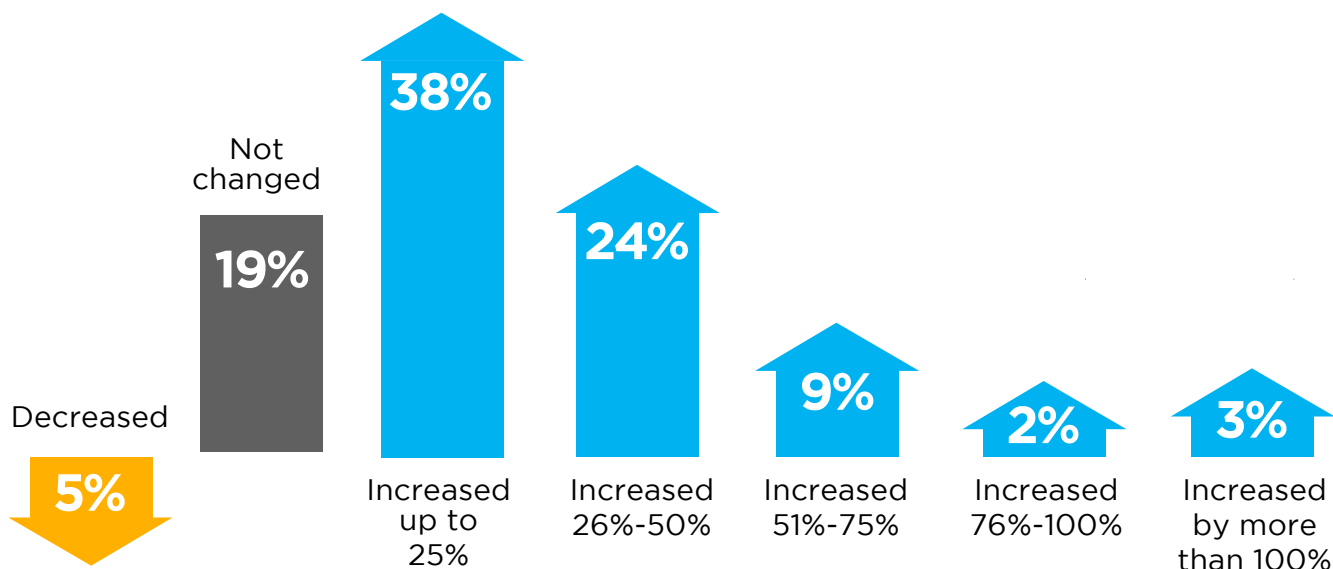
Most organizations experienced an increase in vulnerabilities over the past 12 months (76%). Vulnerabilities increased by up to 25% for over a third of organizations (38%), followed by 24% who saw vulnerabilities increase between 26% and 50%.

▶ **How has the volume of security vulnerabilities changed over the past 12 months for your organization?**



## 76%

of organizations report an increase in the volume of security vulnerabilities during the past year



# INVESTMENT TRENDS

Slightly more organizations expect an increase in investment (44%) for vulnerability management solutions than not (37%). However, despite this expected increase, less than a third of organizations (30%) expect a corresponding increase in staffing – possibly a reflection of the skill shortage facing cybersecurity teams. Budgets for vulnerability testing is also expected to increase for just over a third of organizations (36%).

## ▶ Do you anticipate an increase in **INVESTMENT** for vulnerability management solutions in the next 12 months?



## ▶ Do you anticipate an increase in **STAFF** for vulnerability management in the next 12 months?



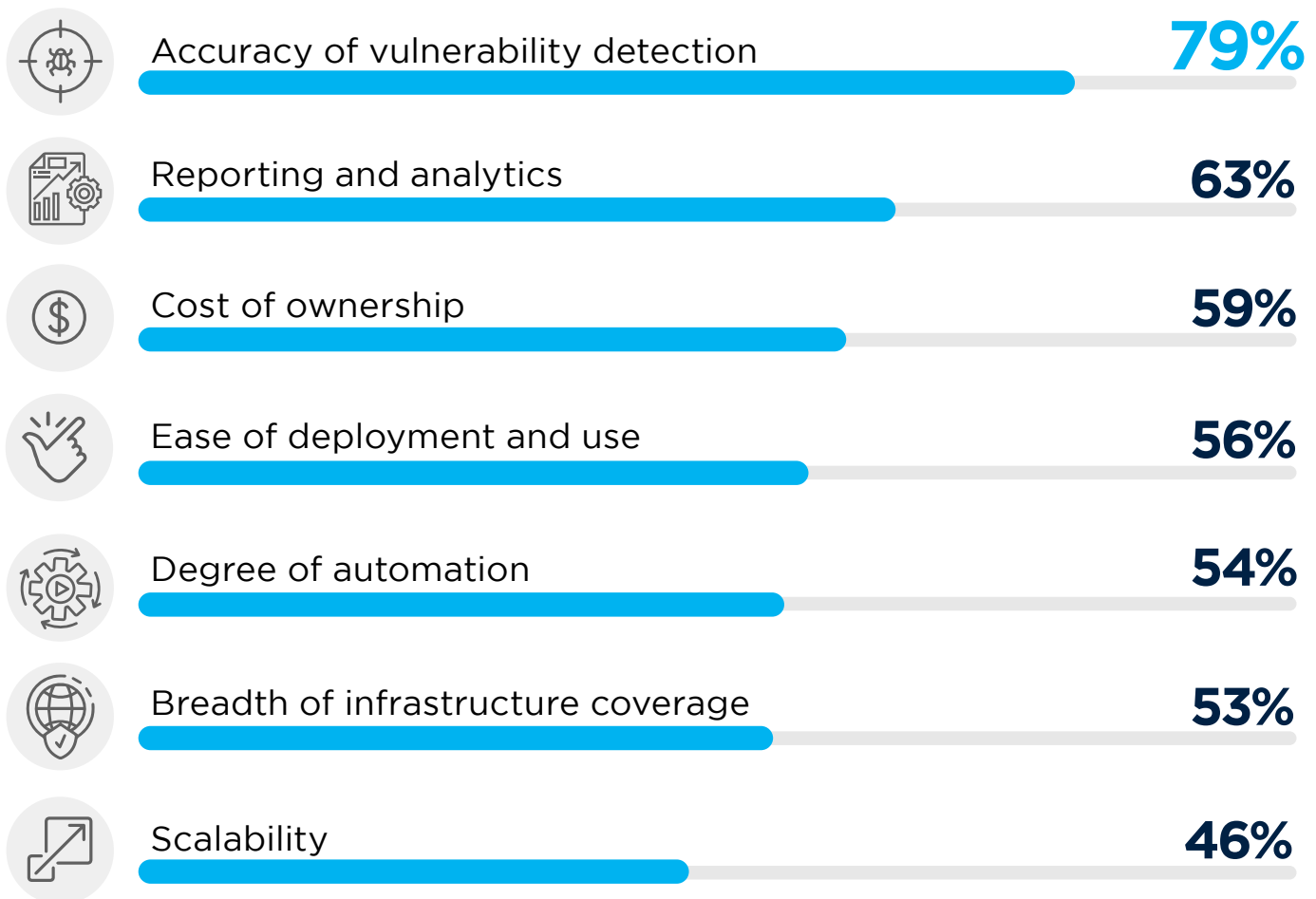
## ▶ Do you anticipate an increase in **BUDGET** for vulnerability testing in the next 12 months?



# SOLUTION EVALUATION CRITERIA

What criteria are most important to organizations when evaluating vulnerability management solutions? Not surprisingly, accuracy of vulnerability detection tops the list at 79% - a core value-add of vulnerability management solutions. This is followed by reporting and analytics features (63%), cost of ownership (59%), and ease of deployment and use (56%).

## ▶ Which purchase criteria are most important to your organization when evaluating a vulnerability management solution?



# CRITICAL SOLUTION FEATURES

Drilling down into vulnerability management capabilities, we asked cybersecurity professionals what solution features they consider most important. Vulnerability assessment (70%) tops the list of features. This is followed by asset discovery (66%), vulnerability scanning (63%), and risk management features (61%).

## ► Which vulnerability management solution features are most important to you?



**70%**

Vulnerability  
assessment



**66%**

Asset  
discovery



**63%**

Vulnerability  
scanning



**61%**

Risk management  
& prioritization



**60%**

Patch  
management



**54%**

Reporting  
and analytics



**50%**

Vulnerability  
intelligence



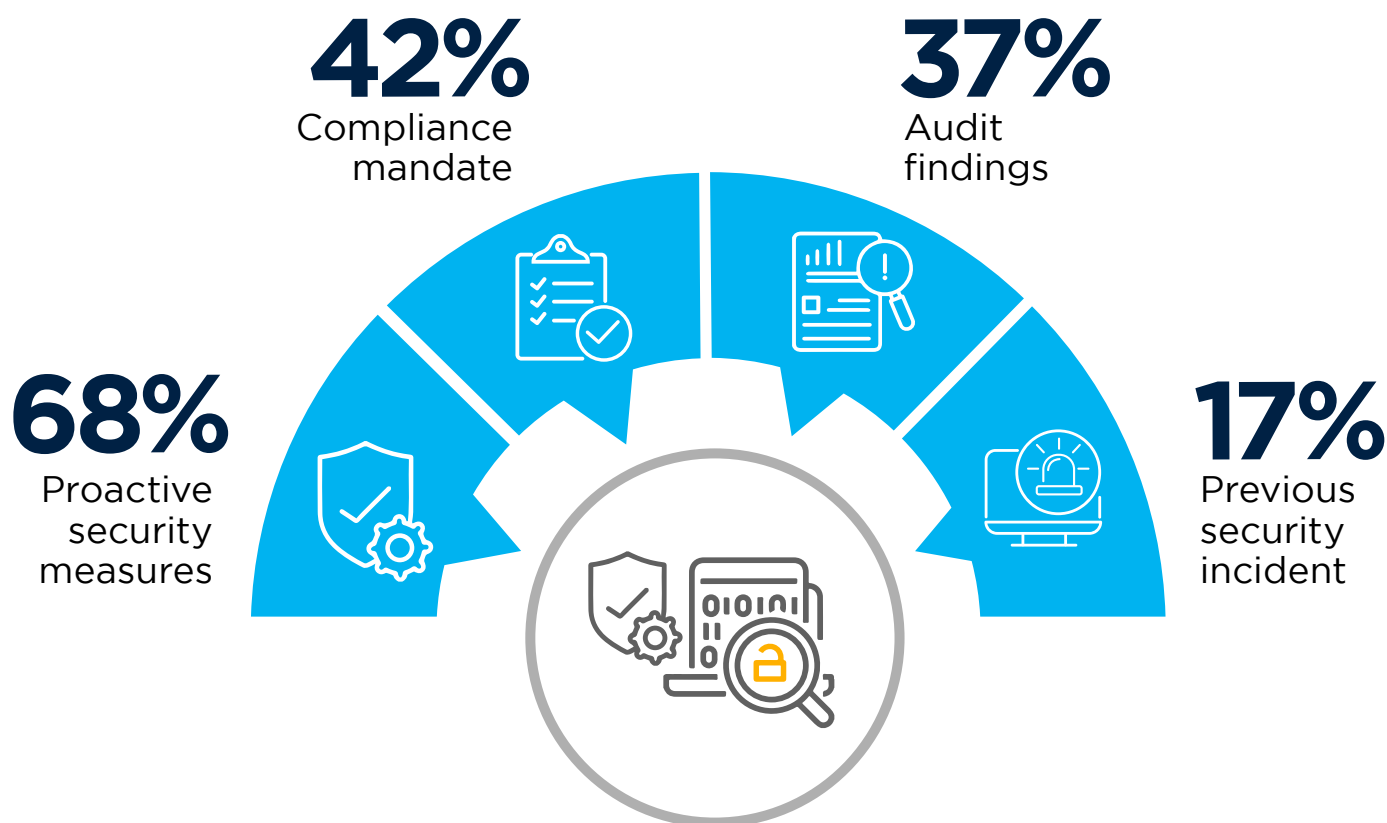
**34%**

Configuration  
monitoring

# PURCHASE TRIGGERS

What motivated organizations to acquire vulnerability management solutions? The vast majority of companies (68%) purchased vulnerability solutions as a proactive security measure, within their cybersecurity program. The next most common driver is compliance (42%), followed by audit findings (37%).

► **What are the main reasons your organization acquired a vulnerability management solution?**

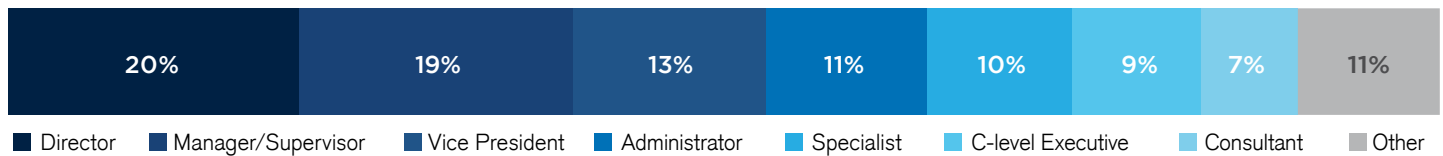


N/A (We have not acquired a vulnerability management solution) 13% | Don't know 8% | Other 2%

# METHODOLOGY & DEMOGRAPHICS

The 2022 Vulnerability Management Report is based on the results of a comprehensive online global survey of 390 cybersecurity professionals, conducted in September 2022, to gain deep insight into the latest trends, key challenges, and solution preferences for vulnerability management. The respondents range from technical executives to managers and IT security practitioners, representing a balanced cross-section of organizations of varying sizes, across multiple industries.

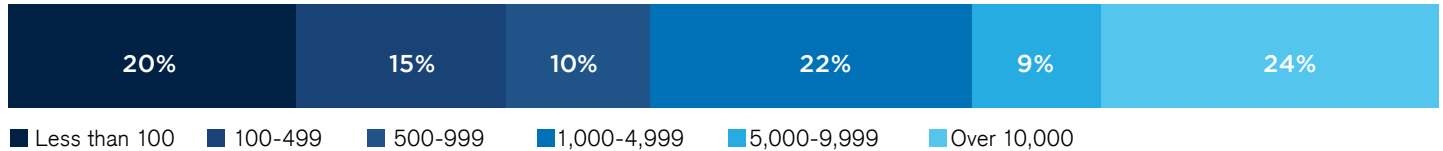
## CAREER LEVEL



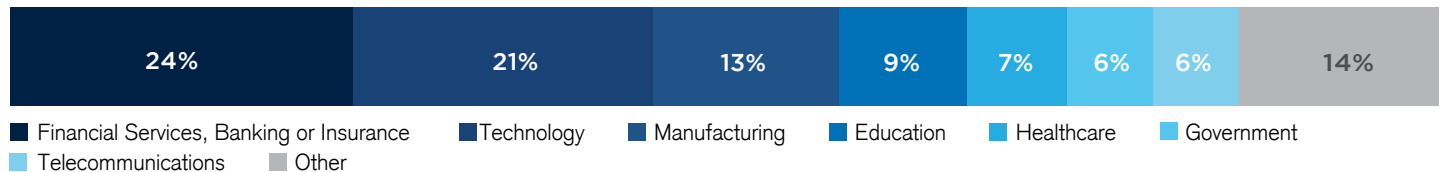
## DEPARTMENT



## COMPANY SIZE



## INDUSTRY





Digital Defense & Beyond Security by HelpSystems are global leaders in vulnerability assessment, management, and compliance – enabling businesses and governments to accurately assess and manage security weaknesses in their networks, applications, industrial systems, and networked software.

[www.helpsystems.com/solutions/data-security/vulnerability-management](http://www.helpsystems.com/solutions/data-security/vulnerability-management)



# Cybersecurity

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